MILITARY SPECIFICATION

BALLISTICS COMPUTER SUBASSEMBLY: 8635274 (ZEROING CONTROL UNIT)

1. SCOPE

1.1 This specification covers one type of electrical mechanical assembly known as Ballistics Computer Subassembly: 8635274, hereinafter referenced as "Zeroing Control Unit" (See 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232 —Parts Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45208 —Inspection System Requirements
MIL-I-45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 —Quality Assurance Terms and Definitions
MIL-STD-202 —Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F8635274 —Ballistics Computer Subassembly: 8635-274 (Zeroing Control Unit)
IEL 8635274 —List of Inspection Documents for Ballistics Computer Subassembly: 8635274 (Zeroing Control Unit)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 **Materials.** Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 **Fabrication.** The zeroing control unit shall be manufactured in accordance with Drawing B863527, and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities.

3.3 **General specifications.** The contractor shall be responsible for adherence to and

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compliance with the following requirements of MIL–F–13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Environmental.

3.4.1 *Operating temperatures.* The zeroing control unit shall meet all the requirements of 3.6 and 3.7 while exposed and thermally stabilized at ambient temperature of −40° and +150° Fahrenheit(F.). Upon return to standard ambient temperature the zeroing control unit shall meet the requirements of 3.4.2.

3.4.2 *Vibration.* The zeroing control unit shall meet the requirements of 3.5 to 3.7.2 inclusive after being vibrated in a vertical plane for five minutes ± 15 seconds at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second.

3.4.3 *Water resistance (sealing).* The interior of the zeroing control unit shall show no evidence of water penetration or other damage after being subjected to a simulated rainfall for a time duration of 15 minutes ±15 seconds.

3.5 Electrical.

3.5.1 *Continuity.* The continuity of the zeroing control unit shall conform to Electrical Schematic Drawing D8635285.

3.5.2 *Dielectric strength.* There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated areas of the zeroing control unit are subjected to an alternating current (ac) root mean square (rms) potential of 300 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than five seconds.

3.5.3 *Insulation resistance.* All insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 10 megohms with 500 volts (dc) applied.

3.6 Performance.

3.6.1 *Zeroing potentiometer output voltages.* The zeroing potentiometer output voltage ratios shall agree with the ratio settings listed in table I for the zeroing knob positions and tolerances shown. No discontinuities in voltage output greater than two steps of potentiometer resolution is allowed between right and left 3.5 mils and up and down 3.5 mils when the 3.5 mil points are approached from the zero mil setting of each knob.

TABLE 1. *Zeroing Control Output Voltage Ratios*

Zeroing knob setting	Ratio setting	Ratio tolerances
Up or left 3 mils	.100000	±.024000
Up or left 2 mils	.233333	±.020000
Up or left 1 mil	.366666	±.016000
Zeroed	.500000	±.004000
Down or right 1 mil	.633333	±.016000
Down or right 2 mils	.766666	±.020000
Down or right 3 mils	.900000	±.024000

3.6.2 *Boresight-normal switch resistance.* Operation of the boresight-normal switches S401 and S402 shall connect the pins of J401 as indicated on schematic Drawing 8635285 with a maximum allowable resistance of 0.5 ohms as measured between pins of J401.

3.6.3 *Circuit breaker function.* Circuit Breaker, CB401 shall function to open the circuit between pins small m and small r of J401 when a dc resistive load of 20 amperes is applied and shall carry 10 amperes continuously without opening the circuit.

3.7 Mechanical.

3.7.1 *Boresight-normal knob operation force.* The maximum force required to pull the boresight-normal knob out from the normal position to the boresight position shall not exceed 12 pounds. The maximum force required to push the boresight-normal knob in from the boresight position to the normal position shall not exceed 12 pounds.

3.7.2 *Zeroing knob operation.* The zeroing knobs when depressed shall rotate freely

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throughout the entire range of scale graduations and shall be capable of engaging the locking segments when released.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or purchase order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL–F–13926 and the Supplementary Quality Assurance Provisions listed on IEL 8635274. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL–STD–105, and the inspection level an dsampling plan specified in table II titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL–P–14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE II. *Classification of Defects*

Use Level II of Table I with Sampling Plan Table II-A of MIL–STD–105

	Requirement	*Test procedure*
Critical: None		
Major: AQL 1.0%		
101. Physical failure (post vibration)	3.4.2	4.5.1.2
102. Continuity	3.5.1	4.5.2
103. Output voltage	3.6.1	4.5.5
104. Switch resistance	3.6.2	4.5.6
105. Circuit breaker function	3.6.3	4.5.7
106. Boresight knob operating force	3.7.1	4.5.8
107. Zeroing knob operation	3.7.2	4.5.9
Minor: None		

Note. The above tests shall be conducted at standard ambient temperature (60° to 90° F.).

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL–STD–105.

4.3 Control tests.

4.3.1 *General.* One zeroing control unit shall be selected at random from each one hundred produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

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Control test	*Requirement*	*Test procedure*
301. Materials	3.1	As per spec. or dwg.
302. Fabrication	3.2	Visual—per dwg.
303. General specification	3.3	Visual—MIL–F–13926

4.3.2 *Environmental and electrical.* Three zeroing control units shall be selected at random from each fifty produced, or from each month's production, whichever occurs first. The control sample shall meet the following requirements and tests:

Control test	*Requirement*	*Test procedure*
304. Operating temperature	3.4.1	4.5.1.1
305. Water resistance	3.4.3	4.5.1.3
306. Dielectric strength	3.5.2	4.5.3
307. Insulation resistance	3.5.3	4.5.4

4.3.3 *Control test failure.* Should any one item of the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of MIL–I–45208 and all applicable requirements specified in MIL–I–45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of MIL–F–13926. Concept, construction, materials, dimensions and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2 *Inspection equipment accuracy requirements.* The devices required in the test methods and procedures herein, to measure the electrical values specified in Section 3, shall provide readings to the accuracy specified below:

Item Number	*Description*
1. Ratio Standard	Ratio range from .00000 to 1.00000 accuracy to ±.01%. Input impedance at 400 cps, 10,000 ohms minimum. Series impedance—10 ohms maximum. Phase shift 0.01° maximum.
2. Oscilloscope	Minimum sensitivity of 20 millivolts, full scale, vertical channel. External synchronization. Input impedance —1 megohm ±10%.
3. A.C. Power Supply	0–5 volts rms, 400 cps ±20 cps, 5% maximum harmonic distortion.
4. D.C. Power Supply	24 volts ±5%, maximum ripple—1.0 volt peak to peak, 30 ampere output.

4.5 Test methods and procedures.

4.5.1 *Environmental.*

4.5.1.1 *Operating temperatures.* This test shall be conducted in conformance with the temperatures specified in 3.4.1. The test equipment shall be in accordance with the "Test Facilities" requirements specified in

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MIL–F–13926. While thermally stabilized at the specified temperatures, the zeroing control unit shall be subjected to the tests specified in 4.5.5 to 4.5.9 inclusive to determine conformance with the requirements referenced therein.

4.5.1.2 *Vibration.* To determine conformance to 3.4.2, the zeroing control unit shall be mounted in its normal position using a suitable mounting apparatus to attach it to a vibrating machine or fixture. Test equipment shall be in accordance with the "Test Facilities" portion of MIL–F–13926 and shall be capable of providing vibration at the amplitude and frequency for the time specified in 3.4.2. After vibration, the zeroing control unit shall be given a visual—tactile examination for damage and loose parts. The zeroing control unit shall then be subjected to the tests specified in 4.5.2 to 4.5.9 inclusive to determine conformance with the respective requirements referenced therein.

4.5.1.3 *Water resistance.* To determine conformance to 3.4.3, this test shall be conducted in accordance with Procedure I of MIL–F–13926 except for the time duration specified in 3.4.3. The zeroing control unit shall show no evidence of water penetration as a result of this test when visually examined.

4.5.2 *Continuity.* The zeroing control unit shall be tested, using standard measuring equipment, to demonstrate continuity of the circuits in accordance with Schematic Drawing D8635307 to determine compliance with 3.5.1.

4.5.3 *Dielectric strength.* The zeroing control unit shall be tested in accordance with Method 301 of MIL–STD–202, and shall meet the requirements specified in 3.5.2.

4.5.4 *Insulation resistance.* Immediately after satisfactorily withstanding the test for dielectric strength, the zeroing control unit shall be given the insulation resistance test in accordance with Method 302, test condition B, of MIL–STD–202, and shall meet the requirements specified in 3.5.3.

4.5.5 *Output voltage.* This test shall be conducted with an ac ratio standard (4.4.2) connected in parallel with the zeroing potentiometers, by connection at pins large E and small a of J401, with a high impedance oscilloscope (4.4.2) connected between the respective zeroing potentiometer slider pins on J401 and the ratio standard slider. Apply 3 volts plus or minus .5 volts ac 400 cps between pins large E and small a of J401. The output voltages shall produce a null at zeroing knob positions and the ratio standard settings of Table I within the tolerances specified. Check for discontinuities in voltage output by rotating each knob from zero position, to their 3.5 mils position right and left, and observing for discontinuities in the oscilloscope display.

4.5.6 *Boresight—normal switch resistance.* This test shall be conducted with an ohmmeter and switching arrangement set up to check resistance between pins small f and large H, small e and small c, small p and small q, large K and small b, large L, and small d, and large J and small g with switches S401 and S402 in the normal position; and between pins large H and large F, small c and large F, small q and large F, small b and large F, small d and large F, small q and large F, and small n and large G with switches S401 and S402 in the boresight position.

4.5.7 *Circuit breaker.* To determine conformance to 3.6.3 apply a 10 ampere dc resistive load to the circuit breaker at pins small r and small m. The circuit breaker shall carry this load without any indication of tripping. Application of a 20 ampere dc resistive load shall trip the circuit breaker.

4.5.8 *Boresight-normal knob force.* To determine conformance to 3.7.1 this test shall be conducted with a "push-pull" force gage with appropriate adapters for attaching to the knob. Attach the gage and adapter to the knob and pull the knob out to its "boresight" position. Observe the force required to accomplish this. Place the gage against the knob and return it to the "normal" position. Again observe the force required. The maximum force required to push or pull the knob

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to its opposite position shall not exceed 12 pounds.

4.5.9 *Zeroing knob operation.* To determine conformance to 3.7.2 depress the elevation zeroing knob and rotate it throughout its range in both directions releasing the knob from time to time and observing its locking capability. Repeat the examination for the remaining 7 knobs. During rotation through all scale range note binding or increased resistance to turning. The knobs shall depress, rotate, and lock as specified in 3.7.2.

5. PREPARATION FOR DELIVERY.

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with MIL–P–14232 and the applicable packaging data sheet. The level of protection shall be as specified in the procurement document (see 6.2).

6. NOTES.

6.1 Intended use. This zeroing control unit is primarily intended as a functional component of the electronic analog Computer, Ballistics M16. The zeroing control unit controls computer power, selects either boresight or normal mode of computer operation and supplies zeroing functions for use in the computer system.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of an applicable level of preservation, packaging and packing (See 5.1).

(c) Applicable packaging data sheet number (see 5.1).

(d) Applicable stock number.

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